



Contact

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Google Scholar: [profile](#)
Doctoral dissertation: [available here](#)

Expertise & Skills

Data engineering and analysis

- Azure, ADF, Databricks, Microsoft Fabric, PySpark, Python, SQL, Snowflake, MATLAB, Crosser, ADE, Docker, git, Azure DevOps

Modeling & Simulation

- FEM (COMSOL)
- MATLAB & Simulink

Other

- MCP Architecture, Azure AI Search
- Miro, Obsidian+Excalidraw
- Bash, PowerShell, MS Office, LaTeX, Markdown
- control engineering, microfluidics

Language

Finnish: Mother tongue
English: C2 (full professional working proficiency)
Swedish: A2
German: A1
Mandarin: A1

Certifications

More information in [Credly](#)

Databricks

Databricks Certified Data Engineer Associate

Agile Data Engine

Agile Data Engine for Data Engineers

AWS

AWS Certified Cloud Practitioner

Azure

Azure AI Fundamentals (AI-900)
Azure Data Fundamentals (DP-900)
Azure Fundamentals (AZ-900)

Scrum

Professional Scrum Master I (PSM I)

ANTTI-JUHANA MÄKI

SENIOR DATA ENGINEER, D.SC. (TECH.)

About Me

Passionate and experienced data guy with a strong mathematical modeling and control engineering background. Building resilient data solutions as a Senior Data Engineer at Siili. Main tools are **Azure** (especially **ADF** and **Azure DevOps**), **Databricks** (PySpark), **Microsoft Fabric**, **Python**, and **SQL**.

Experience

Senior Data Engineer, Siili

2026/02 – Present

Working primarily in Azure, Databricks, Microsoft Fabric, and Snowflake. A growing focus on GenAI topics including MCP architecture, ontology-driven AI systems, and semantic search, emphasizing data quality and governance.

In-house Accelerator Tech Lead & Architect, Solita

2025/09 – 2026/01

Led an in-house agent-based (MCP server) innovation and development team of seven members, serving as Technical Lead, Project Manager, and Scrum Master while collaborating closely with cross-functional stakeholders. Designed the MCP Reference Architecture for internal GenAI accelerators.

Data Engineer, Solita

2023/03 – 2026/01

Worked mainly in Azure. Customers were private companies (energy and Industrial IoT sectors), Football Association of Finland, and Public sector organizations. Also worked with Snowflake, Crosser, git, Docker, ADE, and Microsoft Fabric.

Data Engineer, Futurice

2022/10 – 2023/02

Intensive five months of learning (e.g. 5 certs), and customer work in retail industry, handling their operational data using Azure (especially ADF), PostgreSQL, Databricks, and Python.

Postdoc, Tampere University (TAU)

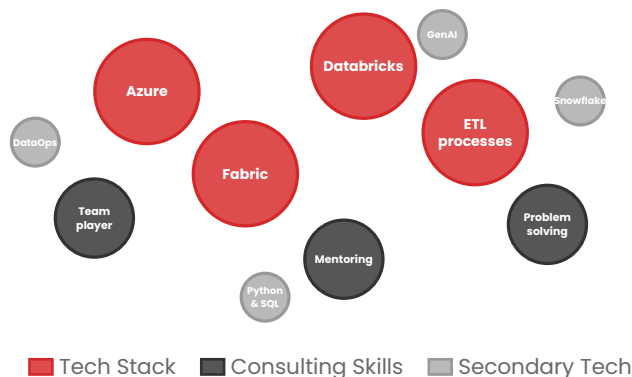
2018/09 – 2022/10

Data analysis & machine learning, control engineering, and mathematical modeling. Additionally, led a group of researchers developing novel in vitro cell culture solutions for vascularized 3D cell co-cultures studies under Professor Pasi Kallio's Micro- and Nanosystems Research Group.

Doctoral Researcher, Tampere University of Technology (TUT)

2011 – 2018

Developed control systems for autonomous cell cultivation platforms, along with extensive mathematical modeling using COMSOL (FEM) and MATLAB/Simulink.



Projects

Data Engineering Services

2026/04 – 2026/08

Private Company, Finland (NDA) · Role: Lead Senior Data Engineer ·

Industry: Food Industry

- Designing and building a scalable **data platform** in Microsoft Fabric
- Lead developer with architectural responsibilities, leading a small development team
- Consolidated a fragmented set of orchestration pipelines into one reliable production pipeline
- Authored the team's technical documentation: architecture guides, naming conventions, and an interactive column-mapping reference

Tech stack: Microsoft Fabric (Lakehouse, Warehouse, Data Pipelines, Notebooks, DAGs, PySpark/Delta Lake), Azure DevOps

Data Engineering Services

2024/03 – 2025/06

Private Company, Finland (NDA) · Role: Data Engineer · Industry: Energy

Sector

- Building and maintaining a scalable **data platform** spanning both **data lakes** and **relational databases**.
- Optimizing **data pipelines** to enhance performance and reliability.
- Ensuring seamless **data integration** across multiple sources for analytics and operations.
- Working with **time-series** data

Tech stack: Databricks, PostgreSQL, Azure, Azure Data Factory, Azure DevOps

Industrial IoT Data Engineering Services

2024/01 – 2024/06

Collaboration with Nokia, Snowflake, and Crosser · Role: Data Engineer and

Data Analyst · Industry: Industry 4.0

- Developing methods to **collect and analyze Industrial IoT (IIoT) data by combining Nokia MX Industrial Edge, Crosser, and Snowflake with Streamlit**. Demo related to Industry 4.0 presented in 2024 Hannover Messe, more info [here](#).

Tech stack: Crosser, Snowflake, Streamlit, ELT

Education

Doctor of Science (Technology)

2018/09

Tampere University of Technology, Tampere, Finland

title: "Modeling and control of microscale cell culture environments", available [here](#).

Master of Science (Technology) "with distinction"

2010/07

Tampere University of Technology, Tampere, Finland

Major: Automation and Control

Student exchange

2006–2007

Hong Kong Polytechnic University

Matriculation Examination

2002

Seinäjoki Upper Secondary School

Merits & Hobbies

Solita Connected Data Clippy Award

2024

- For a person who is always helping and sharing knowledge

Invention disclosures

- TKS0222021 MEA-signaalin analyysityökalu ("MEA analysis tool"), granted 05/2021
- TKS0612020 Pilarivapaa 3D monisolukasvatuschippi ("Barrier-free 3D co-culture chip"), granted 06/2020

Member of the Organizing Committee

Nordic Organ on a Chip Symposium 08/2019

Finnish military rank

Second lieutenant

Hobbies

Football, reading, cross country skiing, orienteering